

Introduction

Autonomous Navigation of Mobile Robots

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Goal of this Module

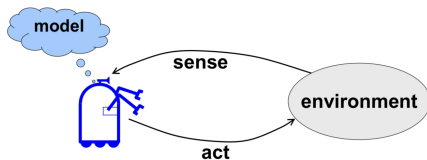
- Provide an overview of problems and approaches in mobile robotics
- Probabilistic reasoning: Dealing with noisy data
- Hands-on experience

Content of this Module

1. Linear Algebra
2. Wheeled Locomotion
3. Sensors
4. Probabilities and Bayes
5. Probabilistic Motion Models
6. Probabilistic Sensor Models
7. Mapping with Known Poses
8. The Kalman Filter
9. The Extended Kalman Filter
10. Discrete Filters
11. The Particle Filter
12. SLAM: Simultaneous Localization and Mapping
13. SLAM: Landmark-based FastSLAM
14. SLAM: Grid-based FastSLAM
15. SLAM: Graph-based SLAM
16. Techniques for 3D Mapping
17. Iterative Closest Points Algorithm
18. Path Planning and Collision Avoidance
19. Multi-Robot Exploration
20. Summary

Autonomous Robot Systems

- perceive their environment and
- generate actions to achieve their goals



Tasks Addressed that Need to be Solved by Robots

- Navigation
- Perception
- Learning
- Cooperation
- Acting
- Interaction
- Manipulation
- Grasping
- Planning
- Reasoning
- etc.

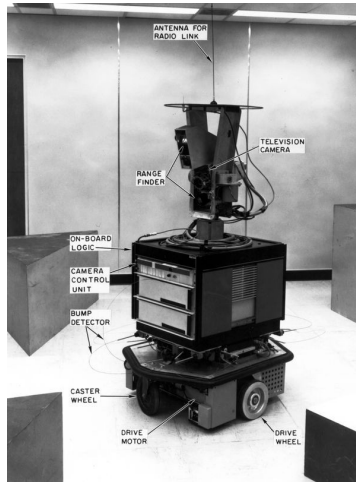
Robotics Yesterday

- Highly repeatable tasks
- Robots bolted to the ground, often “caged”
- Limited to no perception
- Very little “AI”



Robotics Yesterday

Shakey the Robot, 1966



Current Trends in Robotics

Robots are (partly) moving away from factory floors...

- Entertainment, toys
- Personal services
- Medical, surgery
- Industrial automation
- Hazardous environments (mining, harvesting, space, underwater)
- Self-driving cars
- etc.